BUSINESS CASE - TARGET SQL

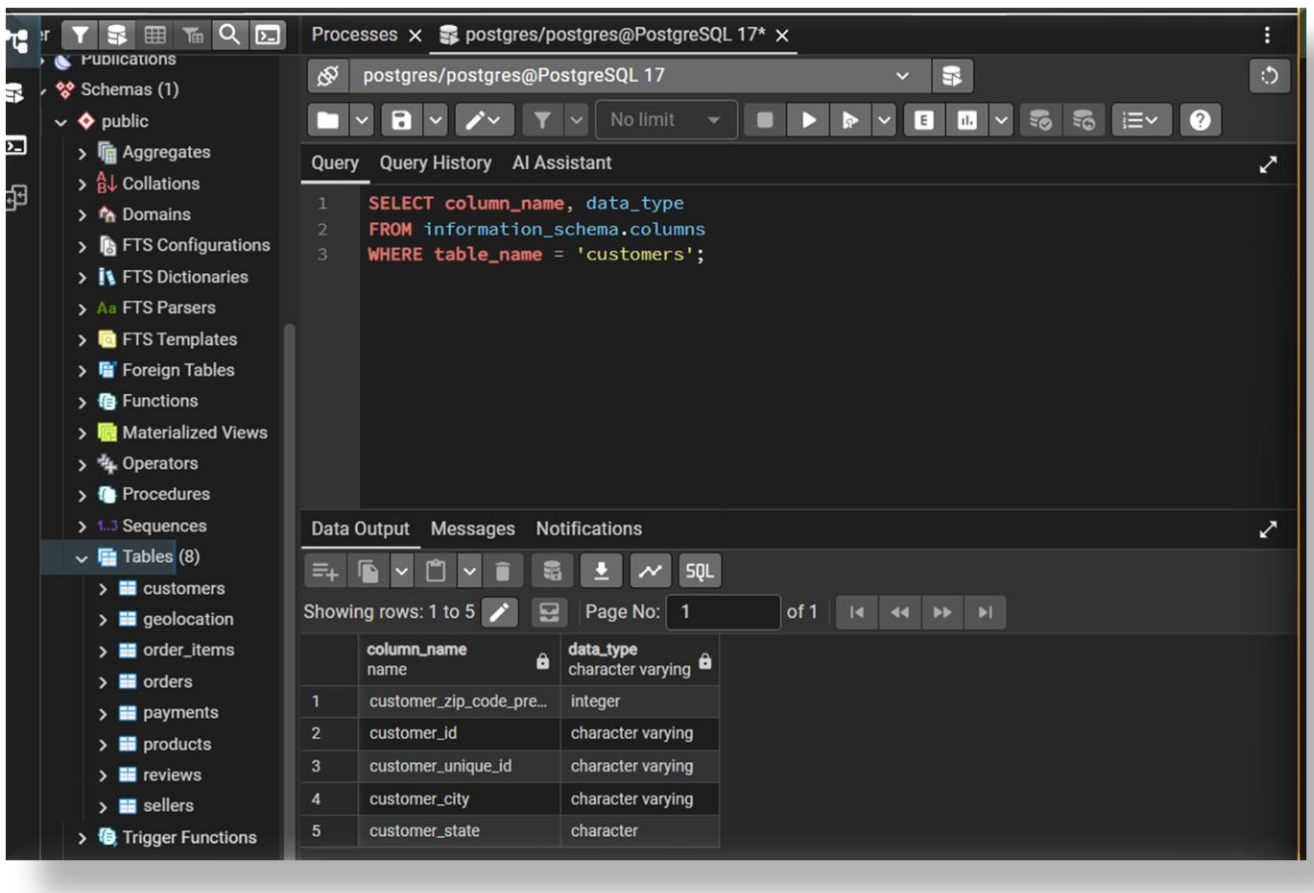
Context: Target Brazil
Target is a leading global retailer known for delivering high-quality products, innovation, and an exceptional guest experience. This case study focuses specifically on **Target's expansion into the Brazilian market**.

Problem Statement
As a Data Analyst at Target, the goal is to evaluate two years of Brazilian transactional data to identify operational bottlenecks and growth opportunities. Specifically, the task is to:

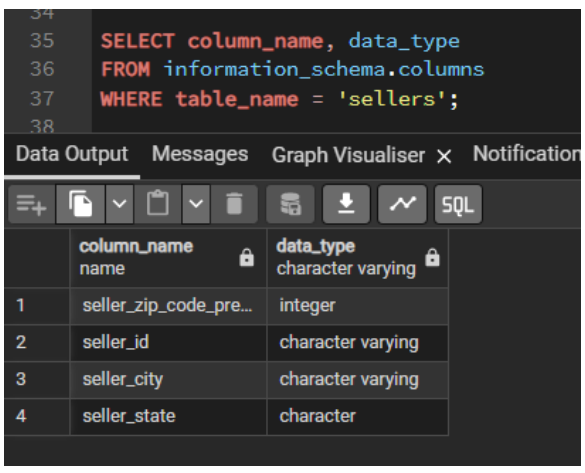
- **Analyse Performance**
- **Audit Logistics**
- **Examine Financials**
- **Provide Solutions**

1. Initial Exploration

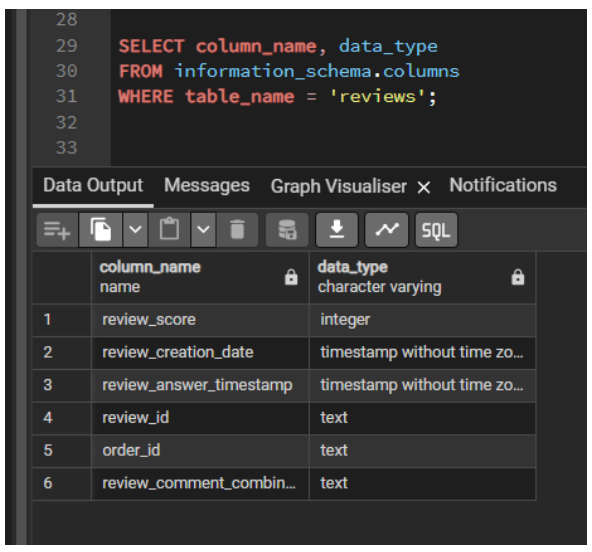
1.1 Data Types of customers Table



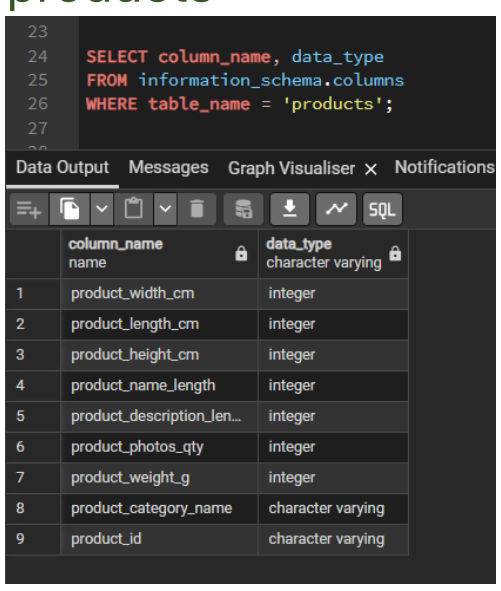
sellers



reviews



products



payments

```sql
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'payments';
```

Data Output | Messages | Graph Visualiser | Notifications

	column_name name	data_type character varying
1	payment_sequential	integer
2	payment_installme...	integer
3	payment_value	numeric
4	order_id	character varying
5	payment_type	character varying

orders

```sql
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'orders';
```

Data Output | Messages | Graph Visualiser | Notifications

	column_name name	data_type character varying
1	order_estimated_delivery_date	timestamp without time zo...
2	order_approved_at	timestamp without time zo...
3	order_delivered_carrier_date	timestamp without time zo...
4	order_delivered_customer_d...	timestamp without time zo...
5	order_purchase_timestamp	timestamp without time zo...
6	customer_id	character varying
7	order_status	character varying
8	order_id	character varying

order Items

```sql
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'order_items';
```

Data Output | Messages | Graph Visualiser | Notificatio

	column_name name	data_type character varying
1	shipping_limit_d...	timestamp without time zo...
2	price	numeric
3	freight_value	numeric
4	order_item_id	integer
5	seller_id	character varying
6	product_id	character varying
7	order_id	character varying

geolocation

```sql
SELECT column_name, data_type
FROM information_schema.columns
WHERE table_name = 'geolocation';
```

Data Output | Messages | Graph Visualiser | Notifications

	column_name name	data_type character varying
1	geolocation_zip_code_pre...	integer
2	geolocation_lat	numeric
3	geolocation_lng	numeric
4	geolocation_city	character varying
5	geolocation_state	character

1.2 Time Range of Orders

Insight: The orders span from **2016-09-04 to 2018-10-17** (approx. 2 years).

```sql
SELECT MIN(order_purchase_timestamp) AS first_order,
       MAX(order_purchase_timestamp) AS last_order
FROM orders;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	first_order timestamp without time zone	last_order timestamp without time zone
1	2016-09-04 21:15:19	2018-10-17 17:30:18

1.3 Count Cities & States of Customers Who Ordered

Insight: Orders originated from apprx **4120 distinct cities** across **27 states** covering wide geographical coverage.

Cities Count

```sql
-- Count distinct customer cities
SELECT COUNT(DISTINCT customer_city) AS distinct_cities
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id;
```

Data Output | Messages | Graph Visualiser | Notifications

	distinct_cities bigint
1	4119

States Count

```sql
-- Count distinct customer states
SELECT COUNT(DISTINCT customer_state) AS distinct_states
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id;
```

Data Output | Messages | Graph Visualiser | Notifications

	distinct_states bigint
1	27

2. In-depth Exploration

2.1 Yearly Growth in Number of Orders

Insight: Orders grew from **329 in 2016 to 45,000 in 2017** then to **54,000 in 2018 .** The e-commerce business expanded rapidly in **2017** and maintained strong growth in **2018**.

```sql
SELECT EXTRACT(YEAR FROM order_purchase_timestamp) AS year,
       COUNT(*) AS total_orders
FROM orders
GROUP BY year
ORDER BY year;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	year numeric	total_orders bigint
1	2016	329
2	2017	45101
3	2018	54011

2.2 Monthly Seasonality

Insight: Peak order volumes occurred during **late Spring** (November 2017: **7,544** orders) and **Summer** (January 2018: **7,269** orders). A secondary peak was observed **early Autumn** (March 2018 : **7211**) remains a strong period for sales.

```sql
SELECT TO_CHAR(order_purchase_timestamp, 'YYYY-MM') AS year_month,
       COUNT(*) AS orders
FROM orders
GROUP BY year_month
ORDER BY year_month;
```

Showing rows: 1 to 25 | Page No: 1 of 1

	year_month text	orders bigint
1	2016-09	4
2	2016-10	324
3	2016-12	1
4	2017-01	800
5	2017-02	1780
6	2017-03	2682
7	2017-04	2404
8	2017-05	3700
9	2017-06	3245
10	2017-07	4026
11	2017-08	4331
12	2017-09	4285
13	2017-10	4631
14	2017-11	7544
15	2017-12	5673
16	2018-01	7269
17	2018-02	6728
18	2018-03	7211
19	2018-04	6939
20	2018-05	6873
21	2018-06	6167
22	2018-07	6292
23	2018-08	6512
24	2018-09	16
25	2018-10	4

Total rows: 25 | Query complete 00:00:00.187 | Rows selected: 1

```sql
SELECT CASE
        WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 5 THEN 'Dawn'
        WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 6 AND 11 THEN 'Morning'
        WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 12 AND 17 THEN 'Afternoon'
        ELSE 'Night'
        END AS time_of_day,
        COUNT(*) AS order_count
        FROM orders
        GROUP BY time_of_day
ORDER BY order_count DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1

	time_of_day text	order_count bigint
1	Afternoon	38361
2	Night	34100
3	Morning	22240
4	Dawn	4740

2.3 Time of Day Distribution

Insight: The majority of orders are placed in the Afternoon (**38,361**) and Night (**34,100**), together accounting for approx **73%** of total orders.Dawn hours we see very less activity.

3. Evolution of E-commerce Orders in Brazil region

Data Output Messages Notifications

	year_month text	customer_state character (2)	orders bigint
539	2018-08	DF	145
540	2018-08	ES	105
541	2018-08	GO	120
542	2018-08	MA	30
543	2018-08	MG	708
544	2018-08	MS	35
545	2018-08	MT	40
546	2018-08	PA	44
547	2018-08	PB	30
548	2018-08	PE	85
549	2018-08	PI	21
550	2018-08	PR	333
551	2018-08	RJ	745
552	2018-08	RN	20
553	2018-08	RO	9
554	2018-08	RS	300
555	2018-08	SC	206
556	2018-08	SE	23
557	2018-08	SP	3253
558	2018-08	TO	13
559	2018-09	MG	4
560	2018-09	RJ	3
561	2018-09	SC	1
562	2018-09	SP	8
563	2018-10	PI	1
564	2018-10	RJ	1
565	2018-10	SP	2

Total rows: 565 Query complete 00:00:00.433

3.1 Month-on-Month Orders per State

```sql
SELECT TO_CHAR(o.order_purchase_timestamp, 'YYYY-MM') AS year_month,
       c.customer_state,
       COUNT(*) AS orders
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY year_month, c.customer_state
ORDER BY year_month, c.customer_state;
```

Insight: SP consistently has the highest monthly order volume, followed by RJ and MG. The state-level trend shows that SP's share of orders increased steadily over time, indicating e-commerce activity in the Southeast.

3.2 Distribution of Customers Across States

Insight: SP holds apprx **42%** of all unique customers, far ahead of other states. This suggests that marketing and logistics efforts should be prioritised in SP, As for the other low performing states , reducing shipping lead times / launching region-specific promotions in these areas could unlock new customer acquisition.

```sql
SELECT customer_state,
       COUNT(DISTINCT customer_unique_id) AS unique_customers
FROM customers
GROUP BY customer_state
ORDER BY unique_customers DESC
LIMIT 10;
```

Data Output Messages Notifications

	customer_state character (2)	unique_customers bigint
1	SP	40302
2	RJ	12384
3	MG	11259
4	RS	5277
5	PR	4882
6	SC	3534
7	BA	3277
8	DF	2075
9	ES	1964
10	GO	1952

4. Impact on Economy

4.1 Percentage Increase in Cost of Orders (Jan–Aug 2017 vs 2018)

Result: Approximately **138%** increase in total order value , indicating that not only did order volume increase,but the total value of those orders escalated dramatically.

```sql
WITH cost_2017 AS (
  SELECT SUM(p.payment_value) AS total_cost
  FROM payments p
  JOIN orders o ON p.order_id = o.order_id
  WHERE o.order_purchase_timestamp BETWEEN '2017-01-01' AND '2017-08-31'
),
cost_2018 AS (
  SELECT SUM(p.payment_value) AS total_cost
  FROM payments p
  JOIN orders o ON p.order_id = o.order_id
  WHERE o.order_purchase_timestamp BETWEEN '2018-01-01' AND '2018-08-31'
)
SELECT ROUND(((c2018.total_cost - c2017.total_cost) / c2017.total_cost) * 100, 2) AS percent_increase
FROM cost_2017 c2017, cost_2018 c2018;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	percent_increase numeric
1	138.53

Data Output Messages Graph Visualiser Notifications

	customer_state character (2)	total_order_price numeric	avg_order_price numeric
1	SP	5202955.05	109.65
2	RJ	1824092.67	125.12
3	MG	1585308.03	120.75
4	RS	750304.02	120.34
5	PR	683083.76	119.00
6	SC	520553.34	124.65
7	BA	511349.99	134.60
8	DF	302603.94	125.77
9	GO	294591.95	126.27
10	ES	275037.31	121.91
11	PE	262788.03	145.51
12	CE	227254.71	153.76
13	PA	178947.81	165.69
14	MT	156453.53	148.30
15	MA	119648.22	145.20
16	MS	116812.64	142.63
17	PB	115268.08	191.48
18	PI	86914.08	160.36
19	RN	83034.98	156.97
20	AL	80314.81	180.89
21	SE	58920.85	153.04
22	TO	49621.74	157.53
23	RO	46140.64	165.97
24	AM	22356.84	135.50
25	AC	15982.95	173.73
26	AP	13474.30	164.32
27	RR	7829.43	150.57

Total rows: 27 Query complete 00:00:00.187

4.2 Total & Average Order Price per State

```sql
SELECT c.customer_state,
       ROUND(SUM(oi.price)::numeric, 2) AS total_order_price,
       ROUND(AVG(oi.price)::numeric, 2) AS avg_order_price
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY total_order_price DESC;
```

Insight: SP remains the primary revenue contributing value of **$5.2M**.
Prices observed in Northern/Remote states (e.g., PE, CE, and PA in your top 15 list) ranging from **$ 145** to **$ 165**
PA (**$ 165**) and CE (**$ 153**) are the true "high average" leaders in the visible data.

Data Output Messages Graph Visualiser Notifications

	customer_state character (2)	total_freight numeric	avg_freight numeric
1	SP	718723.07	15.15
2	RJ	305589.31	20.96
3	MG	270853.46	20.63
4	RS	135522.74	21.74
5	PR	117851.68	20.53
6	BA	100156.68	26.36
7	SC	89660.26	21.47
8	PE	59449.66	32.92
9	GO	53114.98	22.77
10	DF	50625.50	21.04
11	ES	49764.60	22.06
12	CE	48351.59	32.71
13	PA	38699.30	35.83
14	MA	31523.77	38.26
15	MT	29715.43	28.17
16	PB	25719.73	42.72
17	PI	21218.20	39.15
18	MS	19144.03	23.37
19	RN	18860.10	35.65
20	AL	15914.59	35.84
21	SE	14111.47	36.65
22	TO	11732.68	37.25
23	RO	11417.38	41.07
24	AM	5478.89	33.21
25	AC	3686.75	40.07
26	AP	2788.50	34.01
27	RR	2235.19	42.98

Total rows: 27 Query complete 00:00:00.228

4.3 Total & Average Freight Value per State

```
SELECT c.customer_state,
       ROUND(SUM(oi.freight_value)::numeric, 2) AS total_freight,
       ROUND(AVG(oi.freight_value)::numeric, 2) AS avg_freight
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY total_freight DESC;
```

Insight: SP also pays the most in total freight approx **$ 718723** with average shipping cost per order at just **$ 15.15.**
Highest Avg. Freight: RR (**$ 42.98**) and PB (**$ 42.72**) , highlighting logistics cost disparity
Lowest Avg. Freight: SP (**$ 15.15**)
Which also means that the "**cost to serve"** customers in Northern and Northeastern regions is significantly higher, impacting profit margins in those areas.

5. Analysis on Sales, Freight and Delivery Time

5.1 Delivery Time & Estimated vs Actual Difference (Single Query)

```
SELECT order_id,
(order_delivered_customer_date::date - order_purchase_timestamp::date) AS delivery_days,
(order_delivered_customer_date::date - order_estimated_delivery_date::date) AS est_vs_actual_diff
FROM orders
WHERE order_delivered_customer_date IS NOT NULL;
```

Data Output Messages Graph Visualiser Notifications

	order_id [PK] character varying (50)	delivery_days integer	est_vs_actual_diff integer
1	e481f51cbdc54678b7cc49136f2d6af7	8	-8
2	53cdb2fc8bc7dce0b6741e2150273451	14	-6
3	47770eb9100c2d0c44946d9cf07ec65d	9	-18
4	949d5b44dbf5de918fe9c16f97b45f8a	14	-13
5	ad21c59c0840e6cb83a9ceb5573f8159	3	-10
6	a4591c265e18cb1dcee52889e2d8acc3	17	-6
7	6514b8ad8028c9f2cc2374ded245783f	10	-12
8	76c6e866289321a7c93b82b54852dc33	10	-32
9	e69bfb5eb88e0ed6a785585b27e16dbf	18	-7
10	e6ce16cb79ec1d90b1da9085a6118aeb	13	-9
11	34513ce0c4fab462a55830c0989c7edb	6	-20
12	82566a660a982b15fb86e904c8d32918	12	-29
13	5ff96c15d0b717ac6ad1f3d77225a350	5	-9
14	432aaf21d85167c2c86ec9448c4e42cc	11	-9
15	dcb36b511fcac050b97cd5c05de84dc3	14	-13
16	403b97836b0c04a622354cf531062e5f	18	-17
17	116f0b09343b49556bbad5f35bee0cdf	13	-21
18	85ce859fd6dc634de8d2f1e290444043	6	-14
19	83018ec114eee8641c97e08f7b4e926f	13	-15
20	203096f03d82e0dffbc41ebc2e2bcfb7	21	11
21	f848643eec1d69395095eb3840d2051e	4	-10
22	2807d0e504d6d4894d41672727bc139f	5	-13
23	95266dbfb7e20354baba07964dac78d5	18	-26
24	f3e7c359154d965827355f39d6b1fdac	4	-4
25	fbf9ac61453ac646ce8ad9783d7d0af6	29	9
26	acce194856392f074dbf9dada14d8d82	12	-32
27	dd78f560c270f1909639c11b925620ea	9	-7
28	91b2a010e1e45e6ba3d133fa997597be	14	-7

Total rows: 96476 Query complete 00:00:00.134

Insight: Target's logistics maintains an average fulfillment cycle of **12.09** days ,while the forecast error stands at **10.96** days.
Avg. Forecast Error: approx **10.96** days (Actual delivery is earlier than estimated). It means , Target Logistics consistently telling customers that "your package will arrive in approx **22** days," when in reality, trucks arrive in **12** days , where an average customer receives their order approx **11** days sooner than the promised date.

5.2 Top 5 States with Highest & Lowest Average Freight

Highest freight:

```sql
-- Highest ***********
        SELECT c.customer_state,
               ROUND(AVG(oi.freight_value)::numeric, 2) AS avg_freight
        FROM order_items oi
        JOIN orders o ON oi.order_id = o.order_id
        JOIN customers c ON o.customer_id = c.customer_id
        GROUP BY c.customer_state
        ORDER BY avg_freight DESC
LIMIT 5;

-- Lowest *************
        SELECT c.customer_state,
```

Data Output Messages Graph Visualiser Notifications

	customer_state character (2)	avg_freight numeric
1	RR	42.98
2	PB	42.72
3	RO	41.07
4	AC	40.07
5	PI	39.15

Total rows: 5 Query complete 00:00:00.197

Lowest freight:

```sql
-- Lowest *************
        SELECT c.customer_state,
               ROUND(AVG(oi.freight_value)::numeric, 2) AS avg_freight
        FROM order_items oi
        JOIN orders o ON oi.order_id = o.order_id
        JOIN customers c ON o.customer_id = c.customer_id
        GROUP BY c.customer_state
        ORDER BY avg_freight ASC
LIMIT 5;
```

Data Output Messages Graph Visualiser Notifications

	customer_state character (2)	avg_freight numeric
1	SP	15.15
2	PR	20.53
3	MG	20.63
4	RJ	20.96
5	DF	21.04

Total rows: 5 Query complete 00:00:00.185

Highest Freight costs with States in the Northern like RR and PB averaging over **$ 42** per order.
Lowest Freight costs are found in the Southeast led by SP at just **$ 15.15.**
This means that Northern paying twice for shipping compared to Southeast (SP)

5.3 Top 5 States with Highest & Lowest Average Delivery Time

Insight: There is a **20-day** gap between the fastest and slowest states.
SP leads with a optimized average delivery time of just **8.7 days** , whereas RR and AP,
take nearly a full month to arrive.
It also means , Northern customers pay the highest shipping fees **$ 42** and also wait the longest time (Almost a month).

Longest

```sql
-- Highest Average Delivery Time ***********
        SELECT c.customer_state,
               ROUND(AVG( (o.order_delivered_customer_date::date - o.order_purchase_timestamp::date) )::numeric, 2) AS avg_delivery_days
        FROM orders o
        JOIN customers c ON o.customer_id = c.customer_id
        WHERE o.order_delivered_customer_date IS NOT NULL
        GROUP BY c.customer_state
        ORDER BY avg_delivery_days DESC
        LIMIT 5;

-- Lowest Average Delivery Time *************
        SELECT c.customer_state,
```

Data Output Messages Graph Visualiser Notifications Showing rows: 1 to 5 Page No: 1 of 1

	customer_state character (2)	avg_delivery_days numeric
1	RR	29.34
2	AP	27.18
3	AM	26.36
4	AL	24.50
5	PA	23.73

Total rows: 5 Query complete 00:00:00.227 CRLF

Shortest

```sql
-- Lowest Average Delivery Time *************
        SELECT c.customer_state,
               ROUND(AVG( (o.order_delivered_customer_date::date - o.order_purchase_timestamp::date) )::numeric, 2) AS avg_delivery_days
        FROM orders o
        JOIN customers c ON o.customer_id = c.customer_id
        WHERE o.order_delivered_customer_date IS NOT NULL
        GROUP BY c.customer_state
ORDER BY avg_delivery_days ASC
LIMIT 5;
```

Data Output Messages Graph Visualiser Notifications Showing rows: 1 to 5 Page No: 1 of 1

	customer_state character (2)	avg_delivery_days numeric
1	SP	8.70
2	PR	11.94
3	MG	11.95
4	DF	12.90
5	SC	14.91

Total rows: 5 Query complete 00:00:00.138 CRL

5.4 Top 5 States Where Delivery is Fastest Compared to Estimate

Insight: There is a huge "expectation gap" in the Northern regions. While also these states typically have the longest transit times, with estimating delivery windows by **17 to 20 days.** Though customers are receiving their products sooner than the date promised at checkout. "Customer delight " needs to be improved by lowering the promised delivery dates for these regions, which could significantly improve conversion rates.

```sql
SELECT c.customer_state,
       ROUND(AVG( (o.order_delivered_customer_date::date - o.order_estimated_delivery_date::date) )::numeric, 2) AS avg_diff
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
WHERE o.order_delivered_customer_date IS NOT NULL
GROUP BY c.customer_state
ORDER BY avg_diff ASC   -- most negative means earliest relative to estimate
LIMIT 5;
```

Data Output | Messages | Graph Visualiser | Notifications

Showing rows: 1 to 5 Page No:

	customer_state character (2)	avg_diff numeric
1	AC	-20.73
2	RO	-20.10
3	AP	-19.69
4	AM	-19.57
5	RR	-17.29

Total rows: 5 | Query complete 00:00:00.212

6. Analysis Based on Payments

6.1 Month-on-Month Orders by Payment Type

```sql
SELECT
    TO_CHAR(o.order_purchase_timestamp, 'YYYY-MM') AS year_month,
    p.payment_type,
    COUNT(*) AS orders,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY TO_CHAR(o.order_purchase_timestamp, 'YYYY-MM')), 2) AS monthly_share_pct
FROM payments p
JOIN orders o ON p.order_id = o.order_id
GROUP BY 1, 2
ORDER BY 1, 3 DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	year_month text	payment_type character varying (20)	orders bigint	monthly_share_pct numeric
1	2016-09	credit_card	3	100.00
2	2016-10	credit_card	254	74.27
3	2016-10	UPI	63	18.42
4	2016-10	voucher	23	6.73
5	2016-10	debit_card	2	0.58
6	2016-12	credit_card	1	100.00
7	2017-01	credit_card	583	68.59
8	2017-01	UPI	197	23.18
9	2017-01	voucher	61	7.18
10	2017-01	debit_card	9	1.06
11	2017-02	credit_card	1356	71.90
12	2017-02	UPI	398	21.10
13	2017-02	voucher	119	6.31
14	2017-02	debit_card	13	0.69
15	2017-03	credit_card	2016	71.06
16	2017-03	UPI	590	20.80
17	2017-03	voucher	200	7.05
18	2017-03	debit_card	31	1.09
19	2017-04	credit_card	1846	71.80
20	2017-04	UPI	496	19.29
21	2017-04	voucher	202	7.86
22	2017-04	debit_card	27	1.05
23	2017-05	credit_card	2853	72.34
24	2017-05	UPI	772	19.57
25	2017-05	voucher	289	7.33
26	2017-05	debit_card	30	0.76
27	2017-06	credit_card	2463	71.68
28	2017-06	UPI	707	20.58

Total rows: 90 | Query complete 00:00:00.219

Insight:

- **Credit Card** - massive increase in total order volume (growing from 254 orders in Oct 2016 to over 4,900 in Aug 2018),around **74%.**
- **UPI payment** - Observed very slight, decline from **20% down to 17%,** indicates that larger, high-value purchases often required credit card use
- **Debit Card** - For nearly two years, Debit Cards were negligible . However, in June–August 2018, their share suddenly jumped from **0.71% to 4.14%,** possible reason could be new partner ships with new banks, specific promotions etc... Company needs to investigate what caused this shift.
- **Voucher** - final months (Sept–Oct 2018), regular transactions drop off, and Vouchers become the primary "payment" recorded up to **100%).**

Data Output Messages Graph Visualiser Notifications

	payment_installments integer	total_orders bigint
1	0	2
2	1	49060
3	2	12389
4	3	10443
5	4	7088
6	5	5234
7	6	3916
8	7	1623
9	8	4253
10	9	644
11	10	5315
12	11	23
13	12	133
14	13	16
15	14	15
16	15	74
17	16	5
18	17	8
19	18	27
20	20	17
21	21	3
22	22	1
23	23	1
24	24	18

Total rows: 24 Query complete 00:00:00.508

6.2 Orders by Number of Instalments

```sql
SELECT payment_installments,
       COUNT(DISTINCT order_id) AS total_orders
FROM payments
GROUP BY payment_installments
ORDER BY payment_installments;
```

Insight:
credit card usage consistently remained dominant at 75%.

- Full Payment observed at approx 52%.
- 2–4 installments at approx 31.6%
- 10+ installments - A specific segment of customers who only buy when they can spread , Offering "Interest-Free" 10-installment plans could be a powerful for high-ticket items.

7. Actionable Insights & Recommendations

1. Focus on High-Potential States

- Since 42% of our high-value customers are here, we should prioritize rolling out new features, faster delivery, and loyalty rewards in this region.
- Although these states have fewer customers, they spend much more per order ($150+). Focus on selling premium products in these areas to maximize revenue.

2. Optimize Logistics for Remote States

- Opening a small warehouse in the Northeast could significantly cut the current $42 average shipping cost.
- Experiment with free shipping on expensive orders in the Northeast. This encourages customers to buy more at once, making the high travel costs worth it for the company.

3. Capitalize on Seasonal Peaks

- Sales spike from November through March. We should stock up main warehouses by October so we don't run out of inventory during these peak months.
- Most people shop in the afternoon and evening. Offering "Morning-Only" discounts can shift some traffic to earlier hours, helping the warehouse team manage the daily workload more effectively.

4. Promote Alternative Payment Methods

- Credit card processing fees reduce company profits. Offering a small 2-3% discount for using UPI or Debit cards can help protect our margins.
- Many customers prefer 10-month payment plans for expensive items. Promoting high-end electronics or furniture with free installation can help these customers decide to buy.

5. Monitor and Improve Delivery Performance

- We currently deliver 11 days earlier than promised. Updating our checkout to show these faster, more accurate arrival dates will lead to significantly higher sales.
- It takes nearly a month to reach states like RR. Partnering with local delivery companies in these spots can cut wait times, keep customers happy, and reduce long-term churn.

6. Leverage Customer Reviews

- Use "Arrived 10 days early!" stories in advertisements. This builds immediate trust with new shoppers.
- For customers in far-away states, send a "We're almost there" message halfway through the delivery process. This proactive communication makes the long wait feel much shorter.